

Report: Modeling, Dynamics, and Kinematics

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November 14th, 2025

This report is based on the lecture notes from the course Modelling, Dynamics and Kinematics [1].

1 Part 1

Continue the steps after step iv) in the assignment description of Part 1.

1.1

v)

The relative configuration of the i -th and $(i-1)$ -th links is given below.

$$H_i^{i-1}(q) = e^{\tilde{T}_i^{(i-1), (i-1)} q_i} H_i^{i-1}(0) \quad (1)$$

Here, we use the matrix form of the twists.

We apply chain rule to compute the configuration representations of i -th frame in the base frame.

$$H^0_i(q) = \prod_{n=1}^i H_n^{n-1}(q) = H_1^0(q) \cdot H_2^1(q) \cdot \dots \cdot H_i^{i-1}(q) \quad (2)$$

vi)

Now we change the coordinate frame that expresses the twists from the $(i-1)$ -th frame to the base frame.

$$\tilde{T}_i^{0, (i-1)} = H_i^0 \tilde{T}_i^{(i-1), (i-1)} H_0^i \quad (3)$$

To compute the geometric Jacobian, we will transform the matrix form of the twists to the vector forms.

$$J(q) = (\hat{T}_1^{0,0} \quad \hat{T}_2^{0,1} \quad \dots \quad \hat{T}_i^{0, (i-1)}) \quad (4)$$

1.2

Using the inspection method, we calculate the unit twists expressed in the base frame of robotic arms; however, the computation becomes exponentially complex as the degrees of freedom increase. In contrast, using the algorithmic method, we are able to deal with robotic arms with higher degrees of freedom using much simpler calculations. This is because the algorithmic approach solves the robot's kinematics by focusing on the kinematic pairs of each joint, and the relationships between joints and their transformations allow us to compute the complete kinematics of the robotic arm from the base frame to the end-effector frame.

2 Part 2

2.1

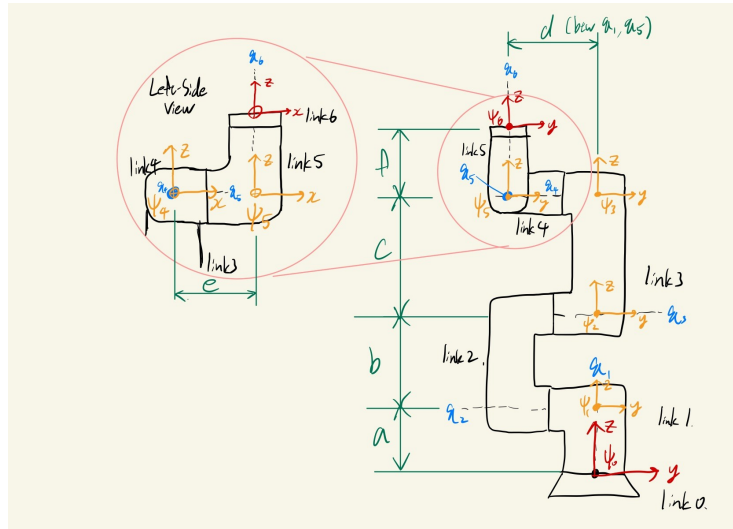


Figure 1: Initial configuration of a robot.

The initial configuration of a robot and the frames are given as Figure 1.

The inputs of the algorithm are computed as follows.

$$\begin{aligned}
\hat{T}_1^{0,0} &= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} & \hat{T}_2^{1,1} &= \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} & \hat{T}_3^{2,2} &= \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \\
\hat{T}_4^{3,3} &= \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} & \hat{T}_5^{4,4} &= \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} & \hat{T}_6^{5,5} &= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \\
H_1^0(0) &= \begin{bmatrix} 0 \\ I_3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & H_2^1(0) &= \begin{bmatrix} 0 \\ I_3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & H_3^2(0) &= \begin{bmatrix} 0 \\ I_3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
H_4^3(0) &= \begin{bmatrix} 0 \\ I_3 & -d \\ 0 & 0 & 0 & 1 \end{bmatrix} & H_5^4(0) &= \begin{bmatrix} e \\ I_3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & H_6^5(0) &= \begin{bmatrix} 0 \\ I_3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
\end{aligned}$$

2.2

The **direct_kinematics.m** contains a function that computing homogeneous matrices of i -th link relative to the base link. The **tilda_twist.m** is used to compute the matrix form of twists. The **robot.m** output the robot as expected, shown in Figure 2 and Figure 3.

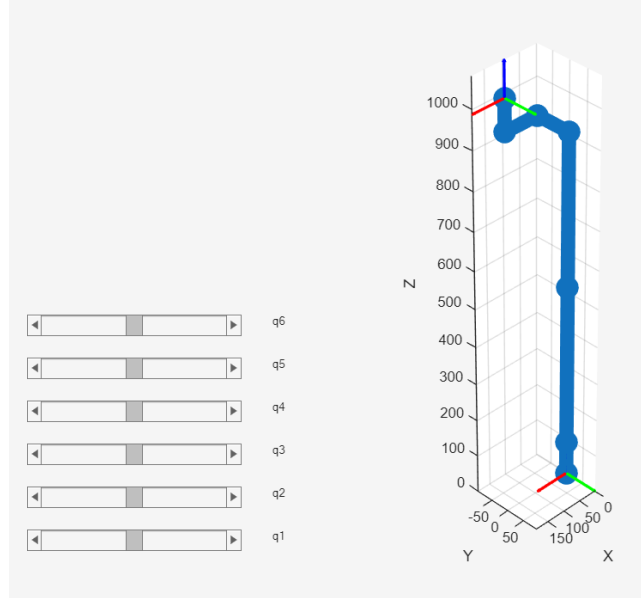


Figure 2: Matlab output of the initial configuration of a robot.

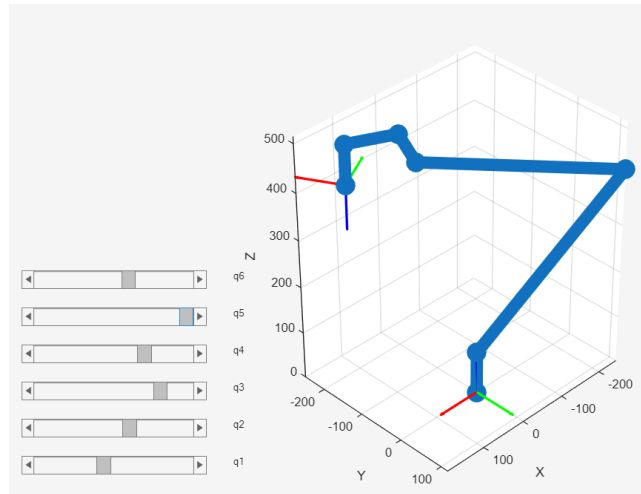


Figure 3: Matlab output of the moved robot

3 Part 3

3.1

The **calculate_dq.m** computes the joint velocities by following steps below.

i)

Given a trajectory of the end-effector frame, the error between the desired and actual positions and the velocity of the end-effector is calculated as described in the assignment.

ii)

The homogeneous matrix H_e^0 is computed using the actual position of the end-effector.

The relative configuration of ψ_0 to ψ_e is given as the inverse of H_e^0 .

$$H_0^e = (H_e^0)^{-1} \quad (5)$$

Given the relative configuration, we can calculate the adjoint matrix of H_0^e , and the $\bar{J}(q)$ is given as the last three rows of the matrix $Ad_{H_0^e} J(q)$.

iii)

By computing the Moore-Penrose pseudo-inverse, the joint velocities are given below.

$$\dot{q} = \bar{J}^\dagger \dot{p}_{ee} \quad (6)$$

3.2

The result of implementation of **robot_control.m** is shown in Figure 4, Figure 5, and Figure 6.

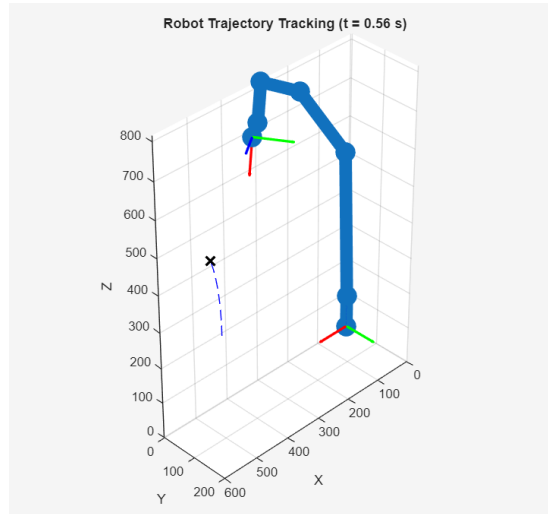


Figure 4: **robot_control.m** execution 1.

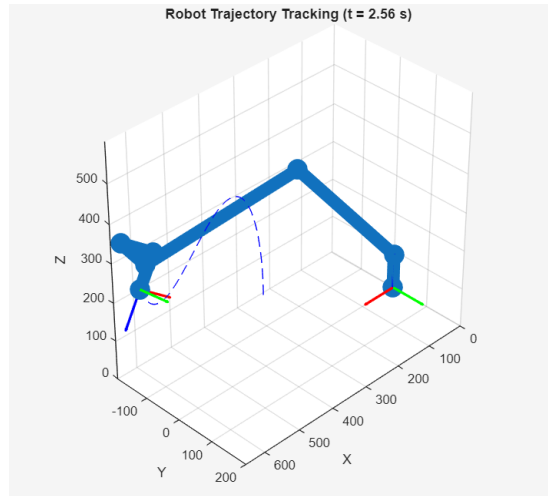


Figure 5: **robot_control.m** execution 2.

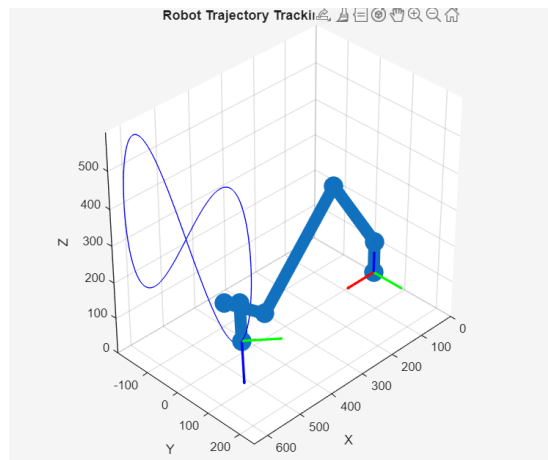


Figure 6: **robot_control.m** execution 3.

References

- [1] Vasily Dunaev. Lecture notes: Modelling, dynamics and kinematics. University of Twente, 2025.